Problem 26.16

Find an expression for the present value of a perpetuity under which a payment of 1 is made at the end of the 1st year, 2 at the end of the 2nd year, increasing until a payment of n is made at the end of the nth year, and thereafter payments are level at n per year forever.

Solution

The payment stream is

$$1, 2, 3, \ldots, n, n, n, \ldots$$

Its present value can be written as the value of an increasing perpetuity $1, 2, 3, \ldots$ minus the excess increases that occur after year n.

The present value of the increasing perpetuity is

$$(Ia)_{\overline{\infty}|} = \frac{1}{id}.$$

Beginning at the end of year $n+1$, the excess payments are

$$1, 2, 3, \ldots$$

Thus, their present value at time 0 is

$$v^n (Ia)_{\overline{\infty}|}.$$

Therefore,

$$PV = (Ia)_{\overline{\infty}|} - v^n (Ia)_{\overline{\infty}|}.$$

Factor the increasing perpetuity value:

$$PV = (Ia)_{\overline{\infty}|}(1 - v^n).$$

Substituting $(Ia)_{\overline{\infty}|} = 1/(id)$ gives

$$PV = \frac{1 - v^n}{id}.$$

Since

$$a_{\overline{n}|} = \frac{1 - v^n}{i},$$

the present value simplifies to

$$\boxed{PV = \frac{a_{\overline{n}|}}{d}}.$$